

Modeling and Calibrating: LLMs Figure out their Decoding Process

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Abstract

An autoregressive language model is trained to predict text, so when it generates text it has an incentive to model its own generation process — including the decoder that samples from its logits. We accumulate evidence that models do exactly this: they infer features of their own decoder (a token blacklist, a top- k truncation, a temperature) from the text in their context, and shift their predictions accordingly. We build a streaming Bayesian filter that estimates the model’s inferred inverse-temperature $\hat{\beta}_t$ online and a corrector that divides a user’s target by $\hat{\beta}_t$. An empirical sweep shows the corrector pins the effective temperature in the “boredom” regime ($\beta^{\text{target}} > 1$) but overshoots in the “confusion” regime ($\beta^{\text{target}} < 1$). Probing what the filter measures sharpens the picture and revises the original framing: on real text the filter is *not* an online version of Guo temperature scaling (it sits ~ 1 nat in $\log \hat{\beta}$ away from the post-hoc MLE, e.g. -0.937 vs $+0.000$ on WikiText), and **Qwen-2.5-3B is already well calibrated on naturalistic English**¹. A controlled sampler-switch experiment falsifies the strong hypothesis that the belief is a fast posterior read off per-token entropy. **The belief that survives is slower and more structural — carried by rank statistics and blacklist signatures, not instantaneous entropy.**² Finally, two metrics for the corrector disagree: closed-loop stability prefers full correction, while invariance on held-out natural text shows full correction is catastrophic unless the filter is given a fading memory ($\gamma < 1$)³. All numbers and tables in this document are generated from the experiments’ data files by build scripts; the prose cites them by name.

1 Introduction

An autoregressive language model emits, at each step, a probability distribution over next tokens given the prefix so far. *Decoding* is the choice of how to sample from that distribution: temperature scaling, top- k , and top- p all act by sharpening or flattening the model’s distribution before drawing the next token. This project starts from a question about what a language model “thinks” about its own decoding process. Because models are trained for text prediction, not generation, they have an incentive to model their own generation process. We hypothesize, and have begun to validate, that *a language model forms a belief about its own decoding parameters by conditioning on the text it has generated so far*. If you sample at a low temperature, the model notices that the recent text is low-entropy and adjusts its output probabilities to match; the low-temperature samples then further condition the model to expect low-temperature text. This is a feedback loop.

¹**[dubious]** The question isn’t whether qwen can handle natural english like wikitext, but whether it can handle other text, like the creative writing on which we found it miscalibrated

²**[dubious]** Rank statistics and entropy are super closely related. This claim rubs me the wrong way. We know *something* is happening, and our first thoughts at nailing it down didn’t work out perfectly, but that doesn’t imply that things we can’t test properly are the right answer. Do you get what I’m saying?

³**[dubious]** Didn’t the stability come from crushing the temperature to zero? That’s not calibration.

Mirostat (Basu et al., 2020) documented a version of this effect and patched it with a control loop on aggregate per-token surprise. That fix cannot adapt to context, because high- or low-entropy text is sometimes exactly what a response calls for. The broader and (to us) more interesting question is to *model* what an LLM believes about its own decoder. This is close to the calibration literature — post-hoc temperature scaling (Guo et al., 2017), its use on language-model surprisal (Liu et al., 2023), and trained universal calibrators (Shen et al., 2024) — and to the algorithms-with-predictions viewpoint (Mitzenmacher and Vassilvitskii, 2020) of the course, in which a possibly-unreliable prediction (here, the model’s inferred decoder) augments a downstream decision. It also connects to recent work on online and high-dimensional calibration (Peng, 2025; Fishelson et al., 2025).

Contributions and revisions. The original proposal framed the streaming estimator as “online temperature scaling converging in the large-data limit to the same global T that Guo’s batch MLE fits.” This write-up reports the experiments we ran to test that framing, and the framing does not fully survive. We find: (i) positive, model-internal evidence that the model infers its decoder (§4); (ii) a corrector that works asymmetrically (§5); (iii) that the filter is *not* Guo MLE on real text and that the well-calibrated-model regime leaves no post-hoc temperature gain (§6); (iv) that the belief is not a fast entropy read-out (§9); and (v) that two natural criteria for tuning the corrector disagree, with memory decay as the load-bearing parameter (§10). The belief-imprint hypothesis is *refined and constrained*, not refuted.

2 Background

For readers new to them, this section glosses the concepts and tools the paper uses, grouped by where they first appear. Each term gets a one- or two-sentence plain-English definition and a footnote pointing to a focused explainer.

Language models and decoding. An **autoregressive language model** emits, at each position, a probability distribution over the next token given all preceding tokens.⁴ That distribution comes from a **softmax**, which turns a vector of real-valued scores (the *logits* ℓ) into positive numbers that sum to one.⁵ **Decoding** is how we sample from it: **temperature** rescales the logits before the softmax (low temperature sharpens the distribution, high temperature flattens it), **top- k** keeps only the k most probable tokens, and **top- p** (nucleus) keeps the smallest set of tokens whose probability mass exceeds p .⁶ We parameterize temperature by its reciprocal, the **inverse temperature** $\beta = 1/T$: multiplying the logits by $\beta > 1$ concentrates the distribution and $\beta < 1$ spreads it out. **Mirostat** (Basu et al., 2020) and post-hoc **temperature scaling** (Guo et al., 2017) are prior methods that tune a single temperature to control surprise or to calibrate confidence.

The filter’s machinery (§3). **Maximum likelihood estimation (MLE)** picks the parameter value that makes the observed data most probable.⁷ The **score** is the gradient of the log-likelihood with respect to the parameter (it is zero at the MLE), and the **Fisher information** is its curvature

⁴E. Voita, “Language Modeling,” *NLP Course For You*: https://lena-voita.github.io/nlp_course/language_modeling.html.

⁵E. Bendersky, “The Softmax function and its derivative”: <https://eli.thegreenplace.net/2016/the-softmax-function-and-its-derivative/>.

⁶M. Labonne, “Decoding Strategies in Large Language Models,” Hugging Face blog: <https://huggingface.co/blog/mlabonne/decoding-strategies>.

⁷StatQuest (J. Starmer), “Maximum Likelihood, Clearly Explained”: <https://www.youtube.com/watch?v=XepXt19YKwc>.

— how sharply the log-likelihood peaks, equivalently how much one observation tells you about the parameter.⁸ **Newton’s method** updates a parameter by $\theta \leftarrow \theta - \text{slope/curvature}$, i.e. score over Fisher.⁹ The **Laplace approximation** replaces a complicated posterior distribution with a Gaussian centred at its peak (the mode), with width set by the Fisher curvature there — this is what turns the belief update into a cheap, closed-form step.¹⁰ Performing that update in the score-over-Fisher metric is exactly **natural gradient** (Fisher scoring): a gradient step preconditioned by the Fisher information.¹¹ The filter places a **log-normal prior** on β (a bell curve on $\log \beta$, so $\beta > 0$ with mean 1) and performs one such Laplace/Newton step per token — a streaming **Bayesian posterior update**.

Evaluation and information (§6, §9). **Cross-entropy** measures how surprised a model is by held-out text (lower is better); **perplexity** is its exponential, and both are reported here in **nats** (natural-log units; 1 nat = $1/\ln 2 \approx 1.44$ bits).¹² The **negative log-likelihood (NLL) surface** is cross-entropy plotted as a function of β ; a key fact in §6 is that it is nearly flat near its minimum. **Entropy** (also in nats) measures how spread-out a distribution is — largest for a uniform distribution and near zero when one token dominates.

Linear algebra and statistics (§4, §5). The **singular value decomposition (SVD)** factorizes a matrix into ranked, orthogonal directions; the share of variance lying *outside* the top direction (the off-axis residual) measures how far the data depart from a single rank-one pattern.¹³ A **bootstrap confidence interval** estimates uncertainty by resampling the data with replacement many times;¹⁴ the **cluster bootstrap** resamples whole groups (here, prompts) rather than individual observations, to respect within-group correlation.¹⁵ A **paired t-test** checks whether the mean of within-pair differences (e.g. the same prompt under two conditions) is non-zero.¹⁶

Closed-loop dynamics (§5, §10). **Teacher forcing** runs a model over fixed ground-truth text rather than over its own samples, isolating its predictions from its generation.¹⁷ The filter’s **memory decay** γ is an exponential moving average: recent tokens count fully while older ones fade, giving an effective window of about $1/(1 - \gamma)$ tokens.¹⁸ A **closed-loop** experiment lets the model condition on its own corrected samples (a feedback loop), whereas a teacher-forced **invariance** test grades the corrector on text it did not generate.

⁸Score and Fisher information, UC Berkeley Stat 210A lecture notes: <https://stat210a.berkeley.edu/fall-2024/reader/score-fisher.html>; and A. Kristiadi, “Fisher Information Matrix”: <https://agustinus.kristia.de/blog/fisher-information/>.

⁹“Taylor Series and Newton’s Method,” suzyahyah.github.io: <https://suzyahyah.github.io/calculus/optimization/2018/04/06/Taylor-Series-Newtons-Method.html>.

¹⁰A. Kristiadi, Laplace-approximation tutorial: <https://agustinus.kristia.de/blog/laplace/>.

¹¹A. Kristiadi, natural-gradient-descent tutorial: <https://agustinus.kristia.de/blog/natural-gradient/>.

¹²“Understanding Evaluation Metrics for Language Models,” *The Gradient*: <https://thegradient.pub/understanding-evaluation-metrics-for-language-models/>.

¹³G. Gundersen, “Singular Value Decomposition as Simply as Possible”: <https://gregorygundersen.com/blog/2018/12/10/svd/>.

¹⁴StatQuest (J. Starmer), “Bootstrapping Main Ideas”: <https://www.youtube.com/watch?v=Xz0x-8-cgaQ>.

¹⁵“The Cluster Bootstrap,” R-bloggers: <https://www.r-bloggers.com/2013/01/the-cluster-bootstrap/>.

¹⁶“The Paired t-Test,” JMP Statistics Knowledge Portal: <https://www.jmp.com/en/statistics-knowledge-portal/t-test/paired-t-test>.

¹⁷J. Brownlee, “What is Teacher Forcing for Recurrent Neural Networks?”: <https://machinelearningmastery.com/teacher-forcing-for-recurrent-neural-networks/>.

¹⁸“Exponentially Weighted Moving Average,” terbe.dev: <https://terbe.dev/blog/posts/exponentially-weighted-moving-average>.

3 Method: a streaming Laplace filter on $\log \beta$

Let $\ell_t \in \mathbb{R}^V$ be the model’s pre-decoding logits at step t and x_t the token actually emitted. We posit a matched-prior *imprint model*: the model behaves as if the natural-reference logits are $\ell_t^* = \ell_t / \hat{\beta}_t$, i.e. the observed logits are the reference scaled by the model’s current belief $\hat{\beta}_t$ about its own inverse temperature. Placing a log-normal prior on β (mean 1, scale σ_0) and doing a per-token Laplace (Newton) update of the posterior on $\eta = \log \beta$ gives a streaming estimator whose update uses the score and Fisher information of the categorical likelihood under $q = \text{softmax}(\ell_t)$:

$$\text{score}_t = \ell_t[x_t] - \mathbb{E}_q[\ell_t], \quad J_t = \frac{1}{\sigma_0^2} + \sum_{\tau \leq t} \text{Var}_q[\ell_\tau].$$

Crucially q , $\mathbb{E}_q[\ell_t]$, and $\text{Var}_q[\ell_t]$ are evaluated at $\beta = 1$ and never re-evaluated at the running $\hat{\beta}_t$: the imprint model is defined precisely so the score and Fisher are $\hat{\beta}$ -independent. This makes the filter a per-token Bayesian update under the imprint model, *not* an iterated Newton solver on a global likelihood — a distinction that becomes important in §6. Three knobs parameterize the belief dynamics: an evidence weight α (per-step step size), a memory decay γ (exponential forgetting; $\gamma = 1$ is no decay, $\gamma < 1$ is a fading window of effective length $1/(1 - \gamma)$), and a correction exponent p .

If the imprint is a single scalar rescaling, dividing the user’s target by the estimate yields a corrector that pins the effective sampling distribution to target:

$$\beta_t^{\text{dec}} = \beta^{\text{target}} / \hat{\beta}_t^p, \quad p = 1 \text{ (full)}, \quad p = 0 \text{ (off)}, \quad p \in (0, 1) \text{ (partial)}.$$

Everything below uses Qwen-2.5-3B (base, fp16). The filter implementation and its unit tests (shift-invariance, batched = serial equivalence, recovery of β_{true} under matched-prior synthetic data) live in `src/decoding.decoding/counter_decode.py`.

3.1 A distribution-free certificate for the grid variant

The Laplace filter is a point estimator; a companion *grid mixture* variant carries worst-case guarantees. Fix a grid $\beta_1 < \dots < \beta_G$ and treat each β_g as an expert forecasting $P_g(x_t) = \text{softmax}(\beta_g \ell_t)[x_t]$ on the *raw* logits (no imprint rescaling — fixed experts are what make the guarantees hold). Let $L_n(\beta) = \sum_{t \leq n} [\text{lse}(\beta \ell_t) - \beta \ell_t[x_t]]$ with $\text{lse} = \log \sum_v \exp(\cdot)$, let π_0 be a prior on the grid, and let the mixture forecast token x_t with probability $\sum_g w_g(t) P_g(x_t)$, where $w_g(t) \propto \pi_0[g] e^{-L_{t-1}(\beta_g)}$ is the Bayes posterior. This is Vovk’s Aggregating Algorithm for log loss (Vovk, 1998; Cesa-Bianchi and Lugosi, 2006), and the following holds for *every* token sequence — no assumption that tokens were sampled from any P_β , or from anything at all.

Lemma 1 (per-sequence certificate). *For any logit sequence $\ell_{1:n}$ and any token sequence $x_{1:n}$: (i) the mixture’s cumulative log loss is $-\ln \sum_g \pi_0[g] e^{-L_n(\beta_g)} \leq L_n(\beta_g) + \ln(1/\pi_0[g])$ for every g ; with uniform π_0 the regret against the best grid temperature in hindsight is at most $\ln G$. (ii) With uniform π_0 the posterior mode equals $\arg \min_g L_n(\beta_g)$, the best-fit (effective) inverse temperature on the grid, exactly. (iii) L_n is convex with $L_n''(\beta) = \sum_t \text{Var}_{q_{\beta,t}}[\ell_t]$ where $q_{\beta,t} = \text{softmax}(\beta \ell_t)$, so for any $c > 0$ the sub-level set $\{\beta : L_n(\beta) \leq \min L_n + c\}$ is an interval containing the sequence MLE; its endpoints, read off the stored grid loss profile, are a data-dependent error bar that needs no distributional assumptions.*

Proof. (i) The mixture’s per-step log probabilities telescope: $\sum_t \ln \sum_g w_g(t) P_g(x_t) = \ln \sum_g \pi_0[g] e^{-L_n(\beta_g)} \geq \ln(\pi_0[g] e^{-L_n(\beta_g)})$ for each g . (ii) Immediate from $w_g(n) \propto e^{-L_n(\beta_g)}$. (iii) Each summand of L_n is a lse of an affine function of β minus an affine function, hence convex; differentiate twice. $\square$

pre-registered hypothesis	outcome	test	verdict
H1 coverage (synthetic, well-spec.)	200/200 cover the MLE	$p = 1.00$	passes
H2 non-vacuousness (Qwen, $\beta=1$)	median width 0.05 (< 0.2 in $\log \beta$)	$p < 10^{-9}$	holds
H3 degradation (peaked vs. natural)	median 0.45 vs. 0.25	$p < 10^{-67}$	holds
H4 transfer (faithful Laplace in bar)	88/96 inside	$p = 0.76$	passes

Table 1: Pre-registered evaluation of the certificate (`paper/grid_certificate_prereg.md`; hypotheses, sample sizes, and thresholds committed before any data were generated). Bars use the default threshold $c = \ln G$ on a 121-point grid spanning $\log \beta \in [-3, 3]$; real streams are Qwen-2.5-3B, 32 prompts, 1000 tokens.

Three honest caveats. First, the certificate is relative to the in-hindsight best-fit β , *not* the dial a counterparty actually set; under misspecified decoders (§4’s blacklist and top- k , the closed loop of §5) no single “true” β describes the stream, and the best-fit value is precisely what “effective temperature” should mean there. Second, the interval in (iii) widens exactly when peaked logits make β unidentifiable ($\text{Var}_q[\ell_t] \rightarrow 0$); forecasting regret and estimation precision degrade in opposite directions. Third, the guarantee attaches to the mixture, not to the Laplace point filter: proper online predictors for logistic-type losses face an exp-concavity obstruction that improper mixture forecasts avoid (Foster et al., 2018), so transfer of the certificate to the Laplace estimates is an empirical claim (tested, with thresholds pre-registered, in `paper/grid_certificate_prereg.md`). A fixed-share variant (Herbster and Warmuth, 1998) tracks a drifting β with regret $\approx m \ln(Gn/m)$ against any m -segment comparator, but the estimation reading (ii)–(iii) is for the static case only.

Table 1 reports the pre-registered evaluation. The bar always covered the continuum MLE on well-specified synthetic streams (200/200); on real $\beta^{\text{target}}=1$ streams the median bar width was 0.05 in $\log \beta$ — the grid resolution floor, i.e. the certificate is far from vacuous where logits carry temperature information — and it widened on peaked-logit streams exactly as the identifiability argument predicts. The transfer check passed: 88/96 re-linearized (faithful) Laplace estimates fell inside the certified interval. An exploratory companion number is more pointed: the original $\beta=1$ -linearized filter of §3 landed inside the bar on only 10/96 of the same streams, an independent, assumption-free confirmation that its estimates are biased away from the sequence MLE once β leaves the neighborhood of 1 — so the certificate extends to the faithful variant’s outputs, not to the original filter’s.

4 The model infers its decoder: positive evidence

Before modeling the belief, we establish that there is one. Two controlled experiments isolate the model’s response to a known decoder manipulation, holding content fixed and comparing treatment to control with cluster-bootstrap confidence intervals.

Blacklist self-prediction. When the k highest-probability tokens are banned (set to zero probability) at sampling time, the model’s own softmax probability on the banned token drops between early and late sequence positions — it learns its blacklist from the absent-token signature. For the rank-1 ban (the token ‘the’), the late-minus-early within-treatment change is $\Delta p_{\text{late-early}} = -0.019$, and the paired late treatment-vs-control difference is -0.016 with 95% CI $[-0.019, -0.013]$, clear of zero (Table 2, Fig. 1).

Table 2: Blacklist self-prediction (run 2026-05-10). CI is the cluster-bootstrap 95% interval on the late treatment-minus-control difference.

banned rank	token	$\Delta p_{\text{late-early}}$	treat.-ctrl (late)	95% CI
1	the	-0.019	-0.016	$[-0.019, -0.013]$
2	,	-0.017	-0.019	$[-0.022, -0.016]$
3	.	-0.017	-0.018	$[-0.021, -0.015]$
4	of	-0.011	-0.004	$[-0.005, -0.002]$
5	to	-0.005	-0.000	$[-0.002, 0.001]$

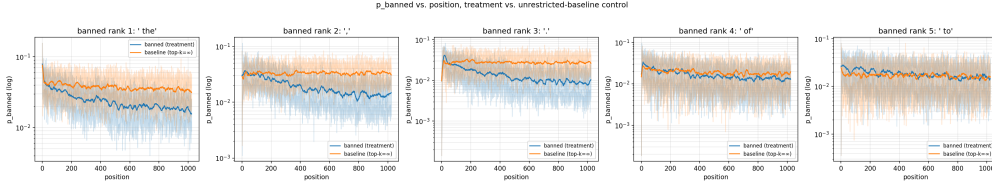


Figure 1: Model probability assigned to the banned token vs. sequence position, per banned rank. Probability mass on a commonly-banned token decays as the model infers the ban from its absence in the context.

Top- k self-prediction. Under top- k sampling, the model’s predicted-rank distribution develops a cliff at rank K that exceeds the corresponding cliff under unrestricted sampling. The treatment-minus-control gap at rank K is +0.475 for $k = 3$ and +0.170 for $k = 5$ (Table 3). Within-treatment concentration grows between early and late positions for finite k and *shrinks* for $k = \infty$ — this is not generic drift but a response to the truncation.

Table 3: Top- k self-prediction ($L = 2048$, run 2026-05-09): the extra cliff at rank K in the predicted-rank distribution, treatment minus control.

k	treat.-ctrl at rank K	95% CI
3	+0.475	$[0.395, 0.567]$
5	+0.170	$[0.150, 0.191]$

These two results are the load-bearing positive evidence: the model demonstrably conditions its predictions on which tokens its decoder did and did not allow.

5 The corrector and concentration dynamics

We next put the filter in a closed loop. Self-generating 200 tokens per prompt at six values of β^{target} , we compare an *uncorrected* arm (fixed $\beta^{\text{dec}} = \beta^{\text{target}}$) against a *corrected* arm ($\beta_t^{\text{dec}} = \beta^{\text{target}} / \hat{\beta}_t$).

Concentration drift. The uncorrected arm sharpens for $\beta^{\text{target}} > 1$ (a “boredom trap”) and flattens for $\beta^{\text{target}} < 1$ (a “confusion trap”); the corrected arm keeps the sampling distribution bounded in both directions (Table 4, Fig. 2). The $\beta^{\text{target}} = 1$ row is a negative control, where the two arms coincide.

Table 4: F0 concentration, late trajectory ($t \in [150, 200]$), corrected vs. uncorrected: max-probability and entropy of the sampling distribution.

β^{target}	corr. max P	unco. max P	corr. H	unco. H
0.5	0.579	0.001	1.94	11.47
0.7	0.521	0.012	2.29	10.48
1.0	0.536	0.534	2.17	2.21
1.3	0.545	0.755	2.11	0.88
1.7	0.602	0.856	1.76	0.44
2.0	0.603	0.891	1.70	0.32

An asymmetry. The corrector is not symmetric. In the boredom regime it pins concentration near its $t = 0$ value, as the theory predicts. In the confusion regime it only partially works: the corrected late max- P_{sample} at $\beta^{\text{target}} = 0.5$ settles at 0.579, well above where it starts — an upward drift the strict (flat-in- t) prediction does not allow. The mechanism is visible in the filter: under uncorrected $\beta^{\text{dec}} = 0.5$, the streaming estimate runs to $\log \hat{\beta} = -4.74$ by $t = 200$ (Table 5), an overshoot. Dividing the target by an overshooting $\hat{\beta}$ produces an over-large β^{dec} , hence the residual sharpening.

Table 5: Streaming $\log \hat{\beta}$ at $t = 200$ under uncorrected decoding (F1.2): sharp decoders push it up, flat decoders push it down, asymmetrically.

β^{dec}	median $\log \hat{\beta}_{t=200}$	IQR
0.5	-4.74	1.25
0.7	-2.71	0.78
1.0	-0.05	0.48
1.3	+0.67	0.35
1.7	+0.98	0.27
2.0	+1.08	0.23

Closed-loop check and multiplicativity. The corrected arm pins $\beta^{\text{eff}} = \beta^{\text{target}}$ by construction (Table 6: the corrected error $E(T)$ is numerically zero), so this is a tautological check that the loop ran; the uncorrected error grows worst at $\beta^{\text{target}} = 0.5$, where $E(T) = 4.74$ nats (a drift of more than two orders of magnitude in β^{eff}). The imprint is *approximately* multiplicative: an SVD of the centered across-decoder logit matrix leaves an off-axis residual fraction that grows to $R = 0.15$ by $t = 200$, so 15% of the across-decoder variation is not explained by a single scalar β .

6 What does the filter actually measure?

The corrector’s confusion-regime failure motivates asking what the filter’s $\hat{\beta}$ is. We test the proposal’s claim — that it is online Guo temperature scaling — directly, and find it false on real text.

No filter pathology under self-sampling. First we rule out the weakest alternative: that the filter drifts on its own. Feeding the filter i.i.d. samples from $\text{softmax}(\ell_t)$ at $\beta^{\text{dec}} = 1$ (the no-imprint

Table 6: Closed-loop error $E(T) = \text{median} |\log \beta^{\text{eff}} - \log \beta^{\text{target}}|$ at $t = 200$ (F3).

β^{target}	$E(T)$ corrected	$E(T)$ uncorrected
0.5	$1.9e-9$	4.74
0.7	$4.5e-9$	2.71
1.0	$0.0e+0$	0.25
1.3	$5.6e-8$	0.67
1.7	$1.3e-8$	0.98
2.0	$1.9e-9$	1.08

null), $\log \hat{\beta}$ flatlines — median -0.070 on WikiText self-generation and -0.052 on WritingPrompts, both within one standard error of zero (Fig. 3). On real Wikipedia continuations the filter agrees with self-sampling (paired $t = -0.52$), but on real WritingPrompts it drifts negative: a paired difference of -0.304 in $\log \beta$ ($t = -3.80$, $p < 10^{-3}$). So the model is well calibrated on Wikipedia and slightly over-confident on Reddit creative writing — a real out-of-domain calibration offset, but an order of magnitude smaller than the drift seen under extreme β -decoding. The closed-loop drift is not a filter artifact.

The streaming filter is not Guo MLE (Exp A). On a proper per-corpus calibration split, the Guo MLE sits essentially at $\log \beta = 0$ on both corpora ($+0.000$ on WikiText, -0.036 on WritingPrompts), confirming Qwen-2.5-3B is already calibrated. The pooled streaming filter on the *same* tokens settles at $\log \hat{\beta} = -0.937$ (WikiText) and -0.802 (WritingPrompts) — about a nat away (Table 7, Fig. 4). The two estimators have *different fixed points*: the filter is a single damped Newton step from the prior mode under the imprint model, accurate near $\beta = 1$ but biased as it walks up the flat NLL surface, whereas the Guo MLE is the global minimizer. Applied as a calibrator the filter *hurts*: on WritingPrompts test, the per-passage streaming $\hat{\beta}$ raises per-token PPL from 19.553 to 24.989 ($\Delta \log \text{PPL} = +0.2522$, $t = +26.46$), while the Guo MLE is statistically indistinguishable from $T = 1$ (paired $t = -0.95$, $p = 0.347$). The proposal’s asymptotic-equivalence claim does not survive; “calibration” and “imprint detection” are distinct targets.

Table 7: Calibration $\hat{\beta}$ per estimator (as $\log \hat{\beta}$; $0 \Leftrightarrow \hat{\beta} = 1$). The pooled streaming filter and the Guo MLE disagree by ≈ 1 nat.

corpus	$\log \hat{\beta}_{\text{stream_med}}$	$\log \hat{\beta}_{\text{stream_pooled}}$	$\log \hat{\beta}_{\text{MLE}}$
WikiText	-0.059	-0.937	+0.000
WritingPrompts	-0.314	-0.802	-0.036

Does the sliding filter track genre (Exp B)? With the calibration framing gone, the remaining question is whether a sliding $\hat{\beta}_t$ tracks content transitions at all. On a sequence alternating WikiText and WritingPrompts chunks, a fading-memory filter ($\gamma = 0.90\text{--}0.95$) sees a small genre signal — the within-chunk mean of $\log \hat{\beta}$ is $+0.054$ higher on WikiText than WritingPrompts, the right sign — but it is dwarfed by the 1.34-nat difference in pre-decoding entropy between the genres (Table 8, Fig. 5). With no decay ($\gamma = 1$) the filter saturates to its pooled-streaming bias ($\log \hat{\beta} \approx -0.46$) and loses all per-context sensitivity. The signal is plausibly content-driven, not a clean read of the decoder.

Table 8: Exp B: mean $\log \hat{\beta}$ per genre (boundary-stripped chunk interiors) and mean pre-decoding entropy per genre, by memory decay γ .

γ	$\log \hat{\beta}$ wiki	$\log \hat{\beta}$ WP	WT-WP	H wiki	H WP
0.90	-0.084	-0.138	+0.054	1.686	3.024
0.95	-0.091	-0.142	+0.051	1.686	3.024
0.98	-0.115	-0.143	+0.028	1.686	3.024
0.99	-0.142	-0.140	-0.002	1.686	3.024
1.00	-0.458	-0.463	+0.004	1.686	3.024

7 Recovering a known decoder temperature

§6 showed the streaming filter is not Guo’s batch MLE on real text — but that comparison collapses the filter to a single scalar and applies it as a *global* temperature, discarding its one real selling point: a *per-token* estimate read causally off the prefix. We test that directly in a controlled setting where the ground truth is known.

A faithful (re-linearised) filter. The production update (§3) evaluates the score and Fisher once at $\beta = 1$ and never re-evaluates, so $\hat{\beta}$ is a single damped Newton step from the prior mode — accurate only near $\beta = 1$ and biased away from it. For the tilted family $P(x | \eta) = \text{softmax}(e^\eta \ell)[x]$ the exact score and expected Fisher in $\eta = \log \beta$ carry chain-rule factors the original drops: $\text{score}_\eta = \hat{\beta}(\ell_x - \mathbb{E}_{q_{\hat{\beta}}}[\ell])$ and $\mathcal{I}_\eta = \hat{\beta}^2 \text{Var}_{q_{\hat{\beta}}}[\ell]$, with $q_{\hat{\beta}} = \text{softmax}(\hat{\beta} \ell)$. The *faithful* filter re-linearises at the running $\hat{\beta}$ each step; over a stream it is a consistent estimator of β , where the original is not.

Setup. For each $T \in \{0.5, 0.7, 1.0, 1.5, 2.0\}$ we seed 64 sequences with a short real prefix (24 tokens, WikiText+WritingPrompts) and sample 232 continuation tokens from $\text{softmax}((1/T) \ell_t)$, so the decoder inverse-temperature is a *known* $\beta = 1/T$. We run both filters over the generated tokens (warmup 20, $\sigma_0 = 0.5$) and, causally, use the pre-update $\hat{\beta}_t$ as a per-token predictive temperature $\tau_t = \hat{\beta}_t$; NLL is computed exactly. A pooled-MLE sanity check recovers $1/T$ at every T (Table 9).

Result. The faithful filter *recovers the imposed temperature* from the prefix alone: its settled $\log \hat{\beta}$ sits within 0.10 of the truth across the range (bias -0.030 at $T=0.7$, -0.091 at $T=2.0$), while the original is badly biased — it overshoots for $\beta > 1$ (bias $+0.478$ at $T=0.7$) and diverges for $\beta < 1$ (bias -4.150 at $T=2.0$). Applied as a causal per-token temperature, the faithful estimate *beats the default $\beta=1$ model* on held-out NLL by 0.108 nats at $T=0.7$ (paired $t = -23.5$) and 0.723 nats at $T=2.0$ ($t = -56.0$), landing within $+0.025$ nats of the oracle (the known-optimal global $1/T$); at $T=1$, where nothing needs correcting, it costs only -0.006 nats. *Honest caveat:* for $T \geq 1.5$ the generated text is near-random (generating-distribution entropy 10.55 and 11.41 nats), so those largest gains calibrate incoherent text; the coherent, sharper-than-default regimes ($T = 0.5, 0.7$) still show clean, significant wins. This is a demonstration that the corrected filter recovers and exploits a decoder temperature from context alone — not a claim that natural text carries such a recoverable temperature (§8 tests exactly that).

Table 9: Controlled recovery: self-text generated at a *known* $\beta = 1/T$. The faithful filter recovers $\log \beta$ (small bias); the original is badly biased. Applied causally per token, the faithful temperature reduces held-out NLL vs $\beta=1$ (ΔNLL in nats, positive = improvement), within the ‘oracle gap’ of the known-optimal global $1/T$. gen. H is the mean entropy of the generating distribution (nats); $T \geq 1.5$ is near-random.

T	$\beta=1/T$	faithful $\log \hat{\beta}$ (bias)	original $\log \hat{\beta}$ (bias)	ΔNLL vs $\beta=1$	oracle gap	gen. H
0.5	2.00	+0.59 (-0.10)	+1.11 (+0.41)	+0.18	+0.009	0.45
0.7	1.43	+0.33 (-0.03)	+0.83 (+0.48)	+0.11	+0.006	1.04
1.0	1.00	-0.01 (-0.01)	-0.12 (-0.12)	-0.01	+0.006	2.80
1.5	0.67	-0.41 (-0.00)	-3.02 (-2.61)	+0.42	+0.005	10.55
2.0	0.50	-0.78 (-0.09)	-4.84 (-4.15)	+0.72	+0.025	11.41

8 Is natural text mis-temperated? A calibration survey

Claim of §7: the filter *can* exploit a decoder temperature when one is present. Does natural text present one? We fit the optimal global temperature (Guo MLE on a train split) per text type and measure its held-out NLL gain over $\beta = 1$, across 9 types spanning in-distribution to far-OOD (60 train / 60 test passages each, length 256; Table 10).

The model is remarkably temperature-calibrated everywhere. Optimal T lies in $[0.91, 1.08]$ across all types, and the best global temperature buys at most +0.0032 nats on any *real* type (only +0.0283 nats even on token-shuffled gibberish, the extreme control). The OOD/atypical types are *statistically* mis-temperated in a sensible direction — mildly overconfident on symbolic and non-Latin text (ABC music $T = 1.03$, $\Delta\text{NLL} +0.0019$, $t = +4.8$; Telugu $T = 1.03$) — but the magnitudes are practically negligible. Difficulty does not imply mis-temperature: Yoruba is the hardest natural text ($\beta=1$ NLL 3.062) yet $T = 1.00$.

Recamán (A005132) is the instructive exception. On this deterministic-but-erratic integer sequence the model is extremely confident and usually right ($\beta=1$ NLL 0.084, entropy 0.14 nats), yet it is the only type that is *underconfident*: optimal $T = 0.91 < 1$, so *sharpening* helps ($t = +11.5$). The predictable structure (delimiters, digit runs) warrants *more* confidence than the model gives, while the rare jumps it cannot anticipate pull the other way. Still, the gain is only +0.0032 nats.

Consequence. With the global-temperature gain $\leq +0.0032$ nats on real text, there is no meaningful temperature miscalibration for a per-token adaptive scheme to exploit: the premise behind an online temperature calibrator on natural text does not hold for this model. Modern LLMs appear temperature-robust even far out of distribution. We confirmed this directly on the two most mis-temperated types (shuffled WikiText and Recamán) with a within-passage cross-validated gate: a per-passage temperature fit on the first half of a passage does *not* beat the single global temperature on the second half (it overfits — on Recamán the per-passage optimum swings wildly across segments but does not transfer), and the causal faithful filter captures no significant gain. The per-passage heterogeneity an adaptive calibrator would need is absent.

Table 10: Calibration survey: held-out NLL gain (nats) of the best global temperature over $\beta=1$, per text type (Guo MLE fit on train, scored on 60 test passages; $*p < 0.05$, $**p < 10^{-3}$), ranked by gain. The model is well temperature-calibrated everywhere (optimal $T \in [0.91, 1.08]$; real-text gains $\leq +0.0032$ nats); shuffled WikiText is an extreme control. $H_{\beta=1}$ is mean predictive entropy (difficulty).

text type	optimal T	$\Delta\text{NLL vs } \beta=1$ (nats)	$\beta=1$ NLL	$H_{\beta=1}$
shuffled WikiText	1.08	+0.0283**	7.20	6.48
Recamán (A005132)	0.91	+0.0032**	0.08	0.14
ABC music	1.03	+0.0019**	2.48	2.34
WritingPrompts	1.04	+0.0011	2.87	2.74
Telugu	1.03	+0.0010**	1.12	1.06
Esperanto	1.03	+0.0007	2.72	2.63
Yoruba	1.00	+0.0001	3.06	3.11
WikiText	1.00	-0.0000	1.96	1.93
Gutenberg (archaic EN)	1.01	-0.0001	1.87	1.85

9 The form of the belief: not a fast entropy read-out

If the model has a belief about β^{dec} , does it show up as pre-decoding entropy that snaps to a new sampler within a few dozen tokens? We self-generate 150 tokens at three β^{dec} and read the pre-decoding entropy $H(\text{softmax}(\ell_t))$ in a late window. The raw effect is enormous: late-window H moves from 1.172 nats at $\beta^{\text{dec}} = 2.0$ to 9.216 nats at $\beta^{\text{dec}} = 0.5$ — a paired spread of 8.04 nats ($t = +69.7$, all 32 prompts ordering the same way; Table 11, Fig. 6).

Table 11: Exp C: late-window ($t \in [80, 150)$) pre-decoding entropy by β^{dec} , with per-prompt paired difference vs. $\beta = 2.0$.

condition	late-window H (nats)	paired diff vs $\beta = 2.0$
$\beta_{\text{dec}} = 0.5$	9.216 ± 0.052	$+8.045 \pm 0.115$ ($t = +69.7$, $n_+ = 32/32$)
$\beta_{\text{dec}} = 1.0$	2.168 ± 0.186	$+0.997$
$\beta_{\text{dec}} = 2.0$	1.172 ± 0.109	—

But it is garbage-in, garbage-out — and the swap control proves it. Natural high-entropy human writing sits far below 9 nats, and the $\beta = 0.5$ generations are token salad within ~ 20 steps: the entropy climbs because each random token makes the context less parseable, not (necessarily) because the model believes the decoder is hot. To separate decoder-belief from content quality, we hold the prefix’s regime fixed and *switch* the sampler at step 80. If the belief were a fast posterior on recent tokens, switching $0.5 \rightarrow 1.0$ should pull late entropy back to the $\beta = 1$ baseline within a handful of tokens. It does not: the `pre05.post10` condition stays essentially flat, from 8.56 (pre-switch) to 8.91 (50 tokens after the switch; Table 12, Fig. 7). Switching $0.5 \rightarrow 2.0$ *does* recover coherence, but only because $\beta = 2$ forces near-top-1 selection that re-anchors the context to in-distribution English — the sampler overwriting the bad prefix, not the model updating a belief.

What this constrains. The decisive comparison falsifies the strong instantiation in which decoder-belief is a fast entropy read-out. It does *not* refute the broader claim — the blacklist and top- k

Table 12: Exp C swap control: median pre-decoding H before the switch and at offsets after it. The decisive row is `pre05_post10`, which does not recover.

condition	$\beta_{\text{pre}} \rightarrow \beta_{\text{post}}$	pre-med	+0	+5	+10	+20	+50
<code>pre05_post05</code>	$0.5 \rightarrow 0.5$ (control)	8.42	9.25	9.21	9.49	9.66	9.47
<code>pre05_post10</code>	$0.5 \rightarrow 1.0$	8.56	9.44	9.10	9.10	8.90	8.91
<code>pre05_post20</code>	$0.5 \rightarrow 2.0$	8.57	9.53	6.83	4.33	1.68	0.33
<code>pre20_post05</code>	$2.0 \rightarrow 0.5$	1.21	0.69	5.20	6.46	8.01	9.06
<code>pre20_post20</code>	$2.0 \rightarrow 2.0$ (control)	1.29	0.64	0.58	1.05	0.30	0.39

experiments (§4) give direct positive evidence, and the WritingPrompts offset (§6) shows the model does respond to which tokens its context contains. The belief operates on slower, more aggregate features — rank statistics, support shape, blacklist signatures — not instantaneous entropy.

10 Belief-update dynamics: closed-loop vs. invariance

Finally we ask which (γ, α, p) best parameterize the belief dynamics, using two complementary criteria. *Closed-loop badness* self-generates under the corrected sampler and penalizes a corrector that fails to track β^{target} . *Belief invariance* teacher-forces the filter on a held-out natural-text continuation after a prefix sampled at varying β^{prefix} , and asks that the corrected per-token cross-entropy be *invariant* to the prefix (a well-calibrated corrector should cancel the prefix imprint and leave the natural distribution untouched).

The two criteria disagree. By closed-loop badness, full correction wins: the default ($\gamma = 1, \alpha = 1, p = 1$) is best (badness 0.011; Table 13). By invariance, full correction at $\gamma = 1$ is catastrophic: cross-entropy on $\beta^{\text{prefix}} = 0.5$ continuations explodes from 3.60 nats at $p = 0$ to 19.44 nats at $p = 1$ (near-uniform over the vocabulary), and the only configurations that stay near the no-correction floor have $\gamma < 1$ (Table 14, Fig. 8).

Table 13: Closed-loop sweep, top 5 by badness. $H@ \beta^{\text{target}}$ and $M@ \beta^{\text{target}}$ are late-window mean entropy and max- P_{sample} .

γ	α	p	$H@0.5$	$H@1$	$H@2$	$M@0.5$	$M@1$	$M@2$	badness
1.00	1.00	1.00	2.45	2.21	1.18	0.53	0.54	0.71	0.011
1.00	1.00	0.75	3.12	1.97	0.92	0.44	0.58	0.75	0.046
1.00	0.50	1.00	3.38	2.27	0.89	0.42	0.52	0.75	0.051
1.00	0.50	0.75	4.56	2.18	0.72	0.31	0.53	0.79	0.089
1.00	1.00	0.50	4.55	2.19	0.66	0.30	0.53	0.80	0.105

Resolving the disagreement. Closed-loop badness is a self-consistency illusion: the filter accumulates a biased $\hat{\beta}$ from the corrected sampler’s own output, then divides the next logits by that same biased $\hat{\beta}$, so the two biases cancel *inside the loop*. “The sampler’s output looks orderly” is necessary but not sufficient for “the filter modeled the belief.” Breaking the loop by teacher-forcing on natural text removes the cancellation, and the bias appears as enormous cross-entropy. Invariance is the more direct criterion, because it grounds the corrector on data the model did not generate. Memory decay $\gamma < 1$ is the load-bearing parameter: it bounds the filter’s worst-case

Table 14: Belief-invariance sweep, top 5 by per-prompt CE variance across β^{prefix} (lower is better). All $p = 0$ configs are identical (correction off) and shown as one “any/any” row.

γ	α	p	CE@ $\beta^{\text{pre}}=0.5$	CE@1	CE@2
<i>any</i>	<i>any</i>	0.00	3.599	3.592	3.631
0.85	0.50	0.25	3.596	3.604	3.639
0.95	0.50	0.25	3.583	3.606	3.632
0.85	1.00	0.25	3.596	3.610	3.643
0.95	1.00	0.25	3.580	3.610	3.631

bias by the fading window’s effective sample size, holding the CE variance to ≤ 0.07 nats² across the entire p range (versus the floor 0.017). The evidence weight α matters little. A reasonable default for off-temperature generation ($\sigma_0 = 0.2$) is moderate decay with partial correction, e.g. ($\gamma = 0.95$, $\alpha = 1.0$, $p = 0.5$); for natural generation ($\beta^{\text{target}} = 1$) no correction is best, since the model is already calibrated.

11 Limitations and conclusion

Limitations. All results use one model (Qwen-2.5-3B), one tokenization, and a small set of corpora; the “Qwen is well-calibrated” finding and the magnitude of the WritingPrompts offset are model- and corpus-specific. The imprint is only approximately multiplicative ($R = 0.15$ off-axis), and no scalar- β correction can cancel non-scalar structure (mode collapse, repetition, vocabulary narrowing); we have not measured how much of the prefix imprint is non-scalar. The invariance test uses only $\beta^{\text{target}} = 1$, the trivially-calibrated case, because there is no natural-text ground truth at other targets. Exp C cannot fully separate decoder-belief from garbage-in/garbage-out, because $\beta = 0.5$ decoheres the prefix faster than any belief update could compensate.

Conclusion. Language models do infer features of their own decoder from context (blacklists, top- k cliffs, an over-confidence offset on out-of-domain text). A streaming Laplace filter that models this as a scalar inverse-temperature belief yields a corrector that works in the boredom regime and overshoots in the confusion regime. The filter is not online Guo temperature scaling — it has a different fixed point and hurts as a calibrator — and the belief it measures is not a fast entropy read-out but something slower and more structural. The right way to tune the corrector is by invariance on held-out natural text, not closed-loop stability, and that criterion makes a fading memory ($\gamma < 1$) essential. The belief-imprint hypothesis is sharpened, not refuted; the open theoretical question is a closed-loop fixed-point analysis of the coupled $(\beta_t^{\text{model}}, \hat{\beta}_t)$ dynamics that would explain the confusion-regime overshoot.

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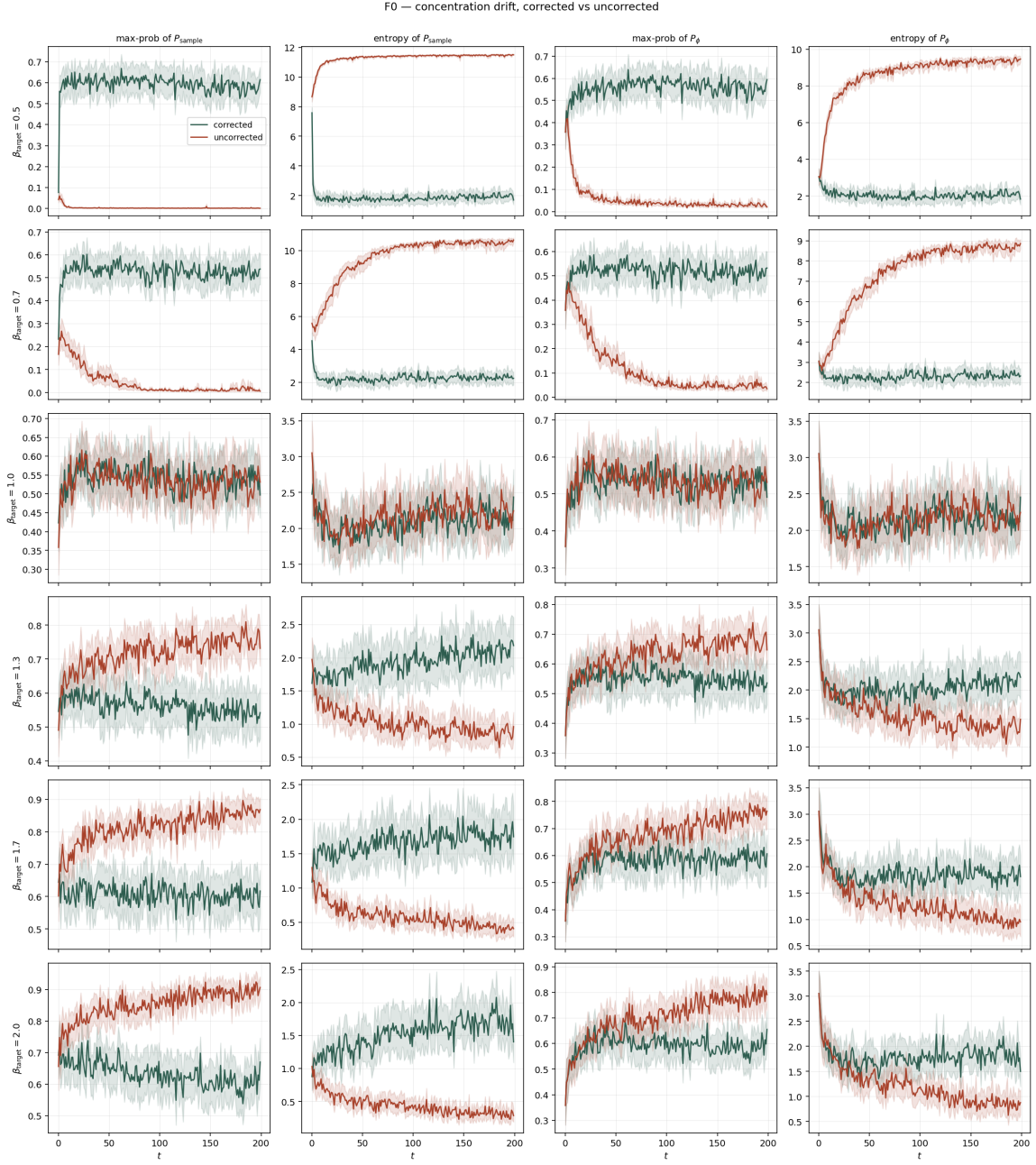


Figure 2: Concentration (max- P and entropy) of the sampling and pre-decoding distributions over t , six β_{target} rows, corrected (green) vs. uncorrected (red). The uncorrected arm runs away in both regimes; the corrected arm stays bounded.

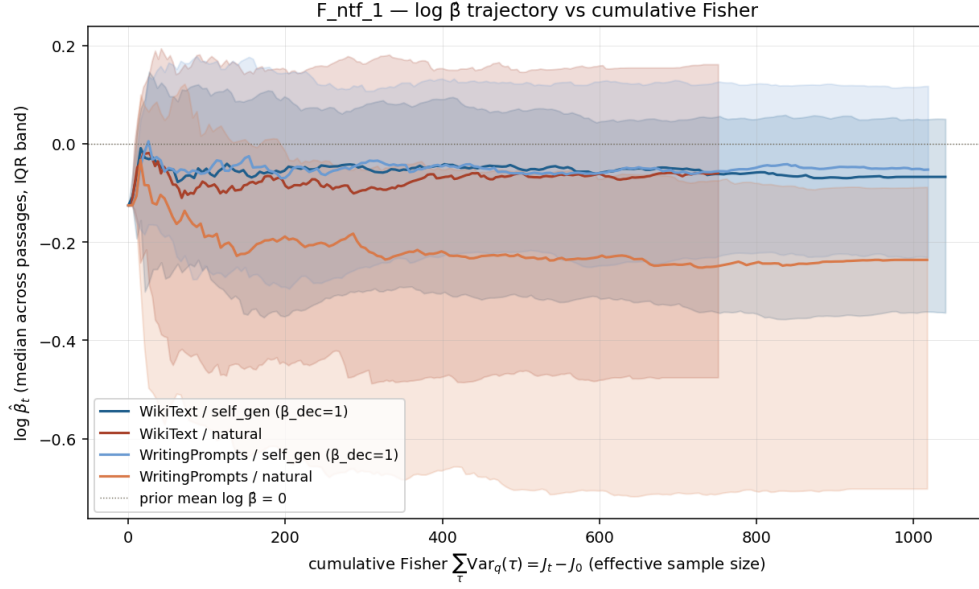


Figure 3: Median $\log \hat{\beta}_t$ vs. cumulative Fisher (the filter’s effective sample size). Self-generation arms flatline at zero; WritingPrompts/natural drifts negative and stays there.

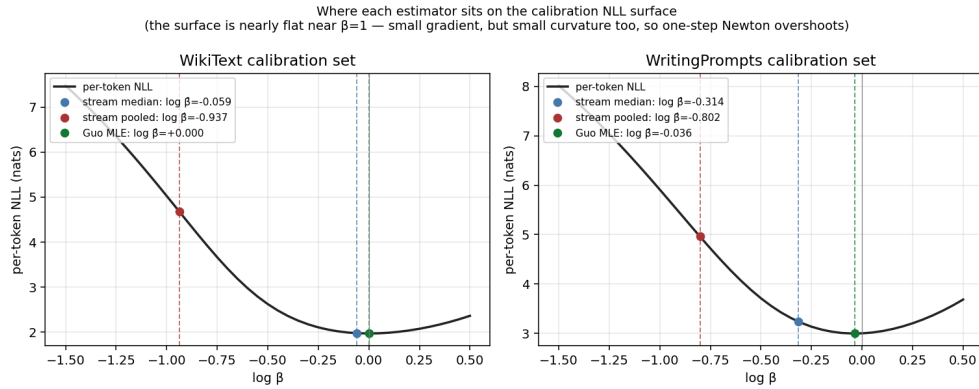


Figure 4: Per-token calibration NLL vs. $\log \beta$. The minimum is at $\log \beta \approx 0$ (Qwen is calibrated); the pooled streaming filter lands far up the steep wall, not at the minimum.

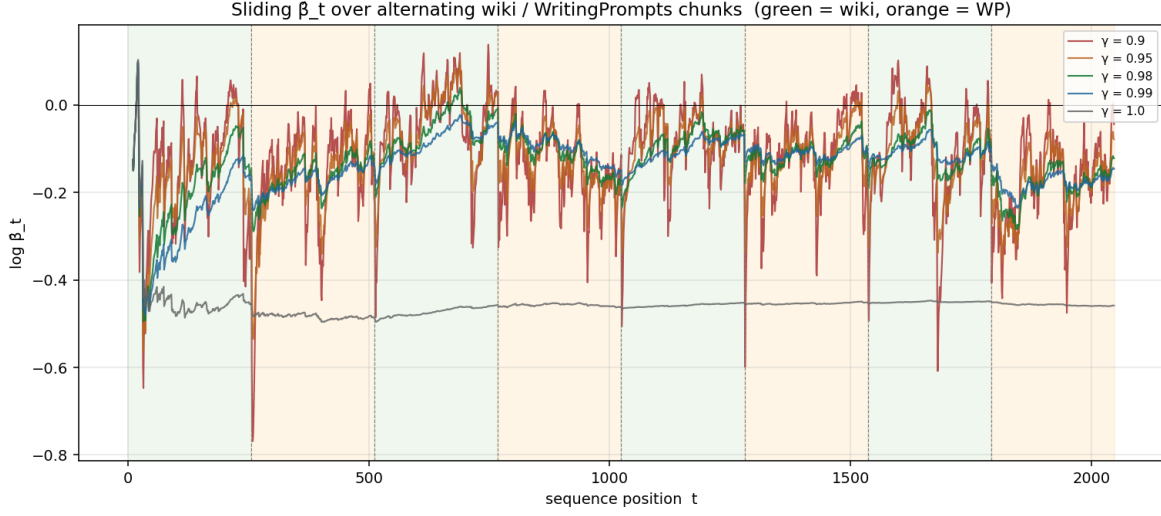


Figure 5: $\log \hat{\beta}_t$ over position; background shading is genre. $\gamma = 1$ (grey) drifts to the pooled bias and stops tracking; damped filters oscillate weakly with genre.

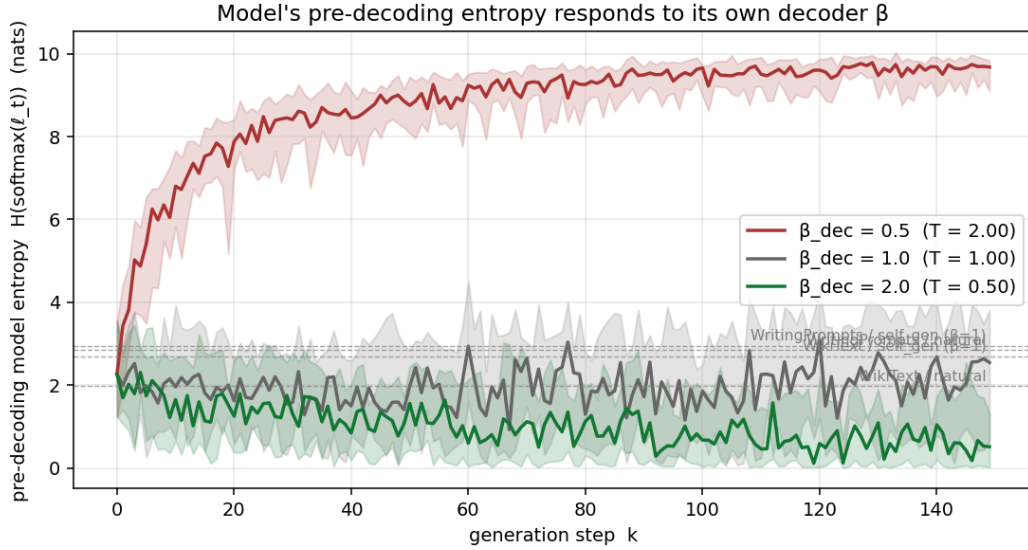


Figure 6: Median pre-decoding $H(t)$ per condition (IQR shaded). Dashed lines mark mean entropies of real natural text; the $\beta = 0.5$ trajectory climbs far past anything natural.

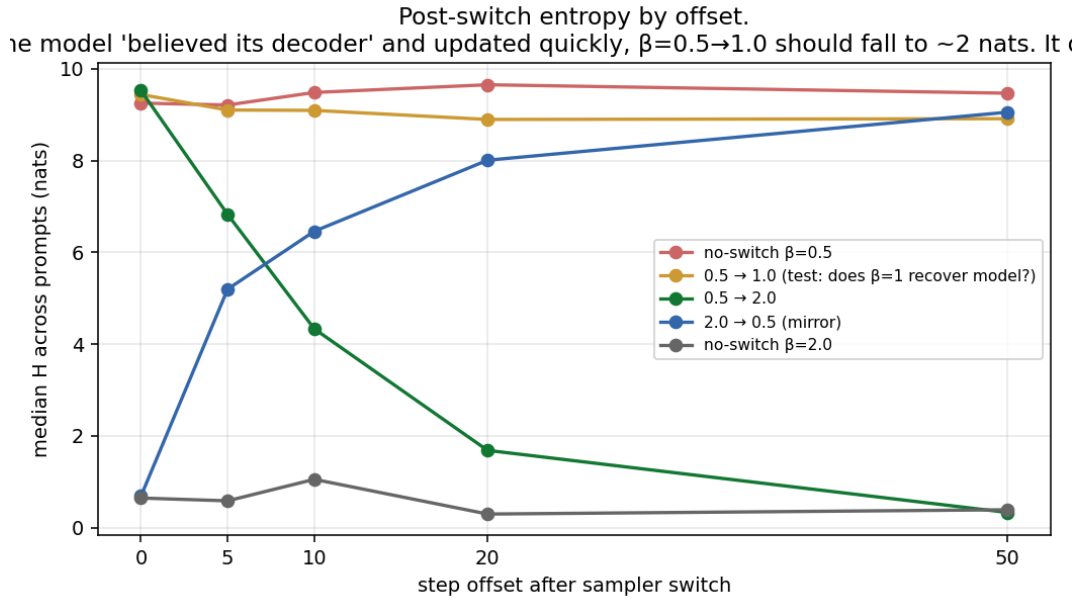


Figure 7: Median pre-decoding H at offsets after the sampler switch. The orange line (pre05_post10, $0.5 \rightarrow 1.0$) does not fall: switching to a neutral sampler does not restore coherence.

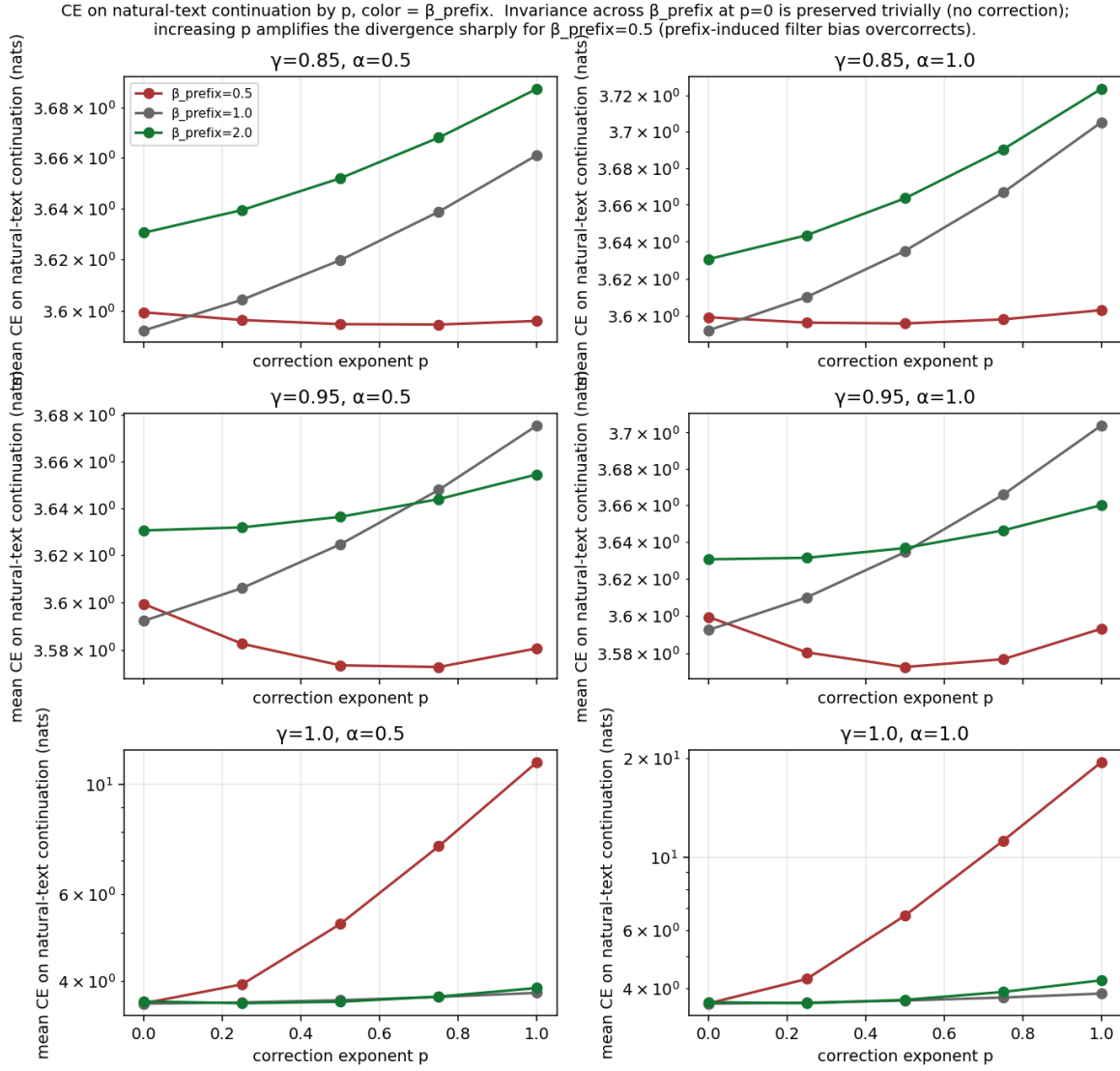


Figure 8: Cross-entropy on the natural-text continuation vs. correction exponent p , colored by β^{prefix} . At $\gamma = 1$ the $\beta = 0.5$ curve grows explosively with p ; at $\gamma < 1$ all curves stay within ≈ 0.1 nat of each other.